The coverage plot

How many features were quantified in each sample, and how far each sample falls short of the dataset as a whole.

AT A GLANCE

Where	Quality control → Coverage
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor
Needs	An uploaded dataset and its annotation. Conditions come from the annotation; sample order follows it.
Two modes	**Summary Type** off draws one bar per sample; **Summary Type** on draws one bar per condition, with the standard deviation and the individual samples over it.
Reference line	The number of features quantified in at least one sample — the ceiling the bars are measured against.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The coverage plot shows how many features were quantified in each sample. It is a quick way to find differences in detection depth before interpreting a comparison. Both the absolute counts and differences between comparable runs matter: equal bars can still represent consistently shallow coverage.

For example, 4,000 measured proteins in one run against about 6,000 in its replicates is worth

investigating. It does not establish that the run failed or that the missing proteins are biologically absent. Check the missing-value pattern, intensity distributions and experimental design alongside it.

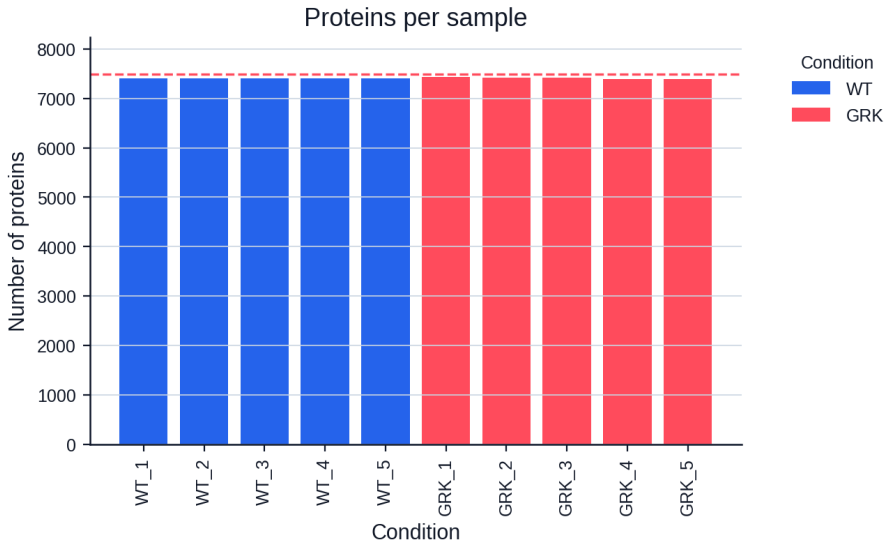


Figure 1.1. Per-sample mode, at protein level, on the example dataset: five WT and five GRK samples, and the reference line across the top.

Reading it

- A One bar, one sample. Its height is the count of non-missing values in that column.
- B The red dashed reference line: the number of features quantified in at least one sample. No bar can exceed it, and the distance to it is what that sample missed.
- C Bar colour: the condition the sample belongs to, taken from the annotation and shared with every other plot in the session.
- D The label under each bar: condition_n, numbered within the condition rather than across the experiment. With **Toggle IDs** on, the original sample identifier is added on a second line.

> EASY TO GET WRONG
>
> A low bar is not by itself a failed sample. The count depends on everything upstream of it — injection amount, gradient length, the search settings — so a bar that is low in every condition alike is usually a property of the run, while a single low bar inside a set of replicates is the one worth investigating. Check it against the missing-value heatmap before excluding anything: a sample that is low but complete in the features it does have behaves very differently from one whose missing values are scattered.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite, phosphoprotein, peptide or precursor. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Summary Type	Off: one bar per sample. On: one bar per condition, showing the mean, the standard deviation and the individual samples.	off	page
Toggle IDs	Adds the original sample identifier under each label.	off	page
Toggle Text	Prints the count on each bar.	off	page
Random Seed	Only in summary mode. Fixes the horizontal jitter of the individual sample points so that the same figure is reproduced exactly.	187	page
Abundance Column	Peptide and precursor levels only. Which quantity column of the report to count.	none until chosen	page
Total Line	Draws the reference line at the number of features quantified in at least one sample. The y axis always leaves the same 10 % headroom: above the line when it is drawn, above the tallest bar when it is not.	on	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size and font. The title is written from the data level: Proteins per sample and its equivalents.	shown	appearance
Legend	Draws the condition legend beside the plot.	shown	appearance
Value labels	Size of the numbers printed on the bars by **Toggle Text**.	9	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Reference line	The line's colour, dash pattern and width. Offered only while **Total Line** is on; whether it is drawn is switched there.	red, dashed	appearance
Condition colours	One colour per condition, shared with every other plot in the session.	the condition palette	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

Every sample column is reduced to a 0/1 matrix: 1 where the value is present, 0 where it is NA. The height of a bar is the column sum of that matrix, so for sample j over m rows

$$\text{count}_j = \sum_{i=1}^{m} \mathbf{1}\left[x_{ij} \neq \text{NA}\right].$$

Only NA counts as missing. A value of zero is a measured zero and is counted as present, which matters for reports that write 0 rather than leaving a cell empty. Samples are renamed condition_n in the order the annotation gives, and the bars are drawn grouped by condition rather than in file order.

SUMMARY MODE

The per-sample counts are computed exactly as above and then aggregated within each condition: the bar is the mean of the counts of the samples of that condition, the error bar is their standard deviation, and each sample is additionally drawn as a point over the bar. With one replicate in a condition there is no spread and no error bar is drawn.

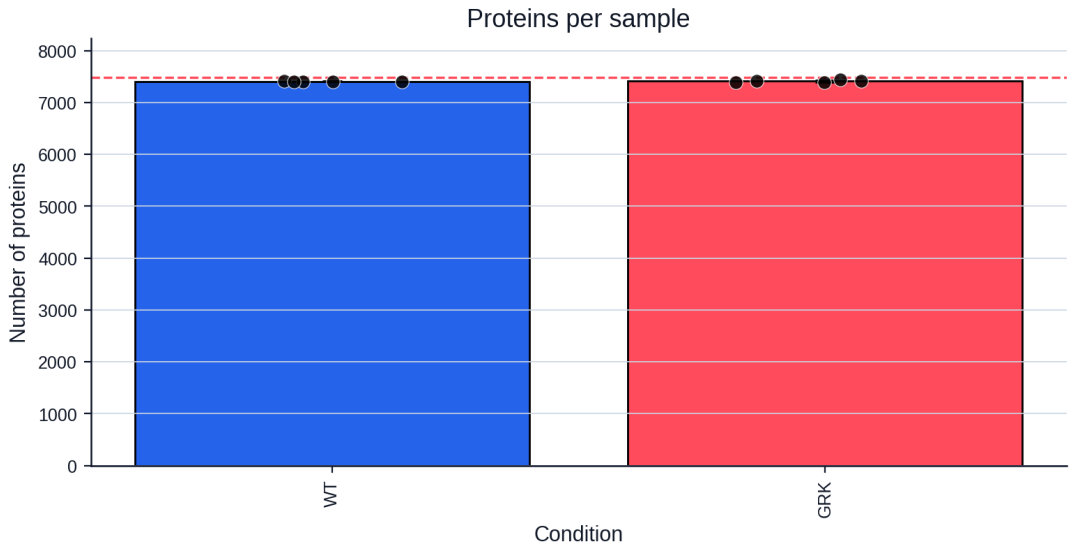


Figure 1.2. Summary mode on the same data: one bar per condition, the standard deviation as an error bar, and the individual samples over it.

THE REFERENCE LINE

The red dashed line is the number of features quantified in *at least one* sample: the size of the union of the bars, not the row count of the matrix. The two differ, because a feature can be identified — a real sequence, a localized site, an assigned protein — and still carry no quantity in any run. Such a row occupies the matrix but appears in no bar, and counting it would put the line above a ceiling no sample could reach. It is the same number as the All samples row of the table below.

NOTE

The same figure drawn in a notebook sits slightly differently: the package function coverage_plot still puts its reference line at the row count of the matrix it is given, so on a dataset with identified but unquantified rows the notebook line is a little higher than the one on screen. The two numbers differ only by the identified-but-unquantified rows, which the table under the plot states on its own row, so a figure from either source can be reconciled with the other.

Edge cases

- A sample with no values at all draws a bar of height zero rather than disappearing from the axis, so the gap is visible.
- A condition with a single replicate gets a bar and a point in summary mode, but no error bar.
- Peptide and precursor levels come from the long-format report rather than a wide matrix, and are counted after the chosen abundance column has been pivoted; until a column is

chosen, no plot is drawn.

- The page always counts the dataset as annotated, before filtering and before imputation. There is no data-source selector here, and that is deliberate: coverage describes what was measured, and counting an imputed matrix would raise every bar to the reference line by construction.

4 The table below the plot

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n as on the plot.
Number	Features quantified in that sample — the height of its bar.
Original_Name	The name the sample carried in the uploaded file. Present when original names are requested.

Two further rows close the table. All samples is the number of features quantified in *at least one* sample, which is the size of the union of the bars rather than their sum. Identified, not quantified appears only when the matrix contains rows that are missing in every sample, and gives their count.

In summary mode the table is replaced by one row per condition, with the mean, minimum, maximum and standard deviation of the per-sample counts of that condition, rounded to two decimals.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = coverage_plot(data, meta, id=False, data_scale="log2")
fig = coverage_plot_summary(data, meta, data_scale="log2")
```

```r
fig <- coverage_plot(data, meta, id = FALSE, data_scale = "log2")
fig <- coverage_plot_summary(data, meta, data_scale = "log2")
```

id=True adds the original sample identifier under each label, as **Toggle IDs** does on the page. The reference line is drawn at the row count of the frame handed in rather than at the number of features quantified anywhere, which is the one difference from the figure on screen.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

Behind this page

Function	What it produces
coverage_plot	Per-sample coverage for wide-format data: protein, phosphosite and phosphoprotein
coverage_plot_pep	Per-sample coverage from the long-format report: peptide and precursor
coverage_plot_summary	The condition-level summary, with spread and individual samples
prepare_coverage_df	The per-sample table under the plot
prepare_coverage_summary_df	The condition-level table